Irreducibility and Galois groups of polynomials

An annotated research bibliography
188 references and resources · Checked through 11 November 2026

A guide to the sources behind the methods: local and global criteria, special polynomial families, random models, specialization, and certified computation.

This collection is selective. Published work, preprints, expositions, corrections, and maintained resources are distinguished. The companion HTML provides full-text search; the BibTeX and JSON files support reuse.

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Reading guide

Scope and conventions

This is a substantial, selective research bibliography on polynomial irreducibility and Galois groups, checked through 11 November 2026. Its centre is univariate polynomials over the rational numbers and number fields. Finite fields, specialization, composition, iteration, and multivariate absolute irreducibility appear as connected branches. A finite guide cannot list every paper in this literature.

Each entry gives a citation, a source link, its publication status, and a short explanation of its use. Where possible, journal versions replace older preprint metadata. Recent preprints retain version information. A few historical items are identified through reliable later references; these are marked. Bibliographic checking and checking a theorem's stated hypotheses do not amount to independently refereeing every proof.

Start with Schinzel for the research literature, Prasolov for a broad collection of techniques, Cox or Milne for explicit Galois theory, and Serre for specialization: [4], [6], [7], [8], [114]. The references are grouped by mathematical purpose; the searchable HTML edition also supports searches by author, title, method, and hypothesis.

The mathematical distinctions that organize the subject

Fix the ground field first. For a separable polynomial f of degree n over a field K, its splitting-field Galois group acts on its n roots. The polynomial is irreducible over K exactly when this action is transitive. Irreducibility alone does not determine the group. Over a finite field, an irreducible degree-n polynomial has cyclic splitting-field group of order n; noncyclic monodromy groups in finite-field papers concern function fields such as F_q(t). See [8], [2], and [147].

Over the rationals, degree-preserving irreducibility modulo one prime proves irreducibility. The converse fails: an irreducible polynomial need not have an irreducible reduction at any prime. More generally, factorizations modulo several primes can rule out every possible degree of a rational factor. At good primes, squarefree factor degrees also give cycle types in the Galois group. See [186], [187], [183], and [125].

For a separable polynomial over a field of characteristic different from 2, its discriminant is a square exactly when the permutation group lies in A_n. Thus a nonsquare discriminant proves the existence of an odd permutation; it does not by itself prove S_n. A common proof strategy establishes A_n containment first, then excludes A_n. A transposition forces S_n in a primitive group; transitivity alone is insufficient. See [37], [188], and [39].

Symmetries change the expected group. For separable reciprocal polynomials of degree 2n, roots occur in inverse pairs, so the natural ambient group is a signed permutation group, subject to the polynomial's additional relations. Remove forced factors before formulating a generic irreducibility claim. Euler polynomials, Bernoulli polynomials, and descent Eulerian polynomials are different families; their similar names do not transfer theorems between them. See [79], [73], and [76].

Choose a proof method

Question or structure	Method to investigate	Starting references
An individual polynomial in Z[x]	Primitive part, modular reductions, shifted Eisenstein, Newton polygons	[186], [12], [13], [32]
Small values at many integers, or a prime value	Schur/Pólya and Cohn-type arguments; track degree and coefficient bounds	[31], [24], [28]
Sparse polynomial or a growing exponent gap	Separate cyclotomic/reciprocal factors; use gap and nonreciprocal-factor theorems	[20], [26], [35]
Explicit Galois group	Frobenius cycle types, discriminants, inertia, primitivity, resolvents	[187], [37], [64], [167]
A parametrized family	Geometric group, Hilbert irreducibility, controlled specialization	[114], [117], [112], [116]
Random coefficients	Specify fixed degree/growing height or growing degree/fixed law; then use the appropriate sieve or mixing theorem	[86], [91], [94]
Iterated polynomials	Capelli-type irreducibility, critical orbits, ramification, tree actions	[123], [124], [121], [126]
Finite-field construction	Irreducibility tests, explicit constructions, prescribed coefficients	[153], [138], [139], [143]
A certified computation	Exact factorization and certified subgroup descent; retain software version and certificates	[155], [158], [167], [174]

Six connected reading routes

1. Classical irreducibility. Read the irreducibility chapter of Prasolov and Conrad's basic notes, then Dumas and Perron. Continue with the Schur/Pólya survey and Schinzel's monograph. This route explains why different criteria work and when a transformation of the polynomial makes them applicable. [6], [186], [12], [13], [31], [4].

2. From arithmetic to S_n or A_n. Start with factorization modulo primes, then group recognition and the discriminant. Study Coleman and Hajir for substantial examples of local arithmetic producing large groups, and Fieker-Klüners for the algorithmic counterpart. [187], [188], [37], [53], [64], [167].

3. Special polynomial families. Begin with Selmer's trinomials and Schur's truncated exponentials; compare Nart-Vila/Osada, Coleman, generalized Laguerre, Bessel, and Fekete polynomials. For each family record the normalization, forced factors, parameter range, and source of the group-theoretic step. [44], [46], [47], [50], [53], [62], [63], [77].

4. Random polynomials. Keep two reading chains: van der Waerden to Gallagher to Bhargava for fixed degree and growing height; Odlyzko-Poonen to Konyagin to Breuillard-Varjú to Bary-Soroker-Koukoulopoulos-Kozma for growing degree. Read the 2026 preprints after these

foundations and check their exact distributions and conclusions. [80], [82], [86], [87], [88], [91], [94], [100], [99].

5. Specialization and inverse Galois theory. Use Serre as the entry point, Fried-Jarden as the reference, and Malle-Matzat or Jensen-Ledet-Yui for realization and generic polynomials. Add quantitative Hilbert irreducibility and effective Chebotarev when a proof needs bounds rather than qualitative existence. [114], [117], [120], [119], [41], [40], [43].

6. Composition, iteration, and monodromy. Read Ritt and modern decomposition theory before Odoni and the arboreal survey. Distinguish generic finite-level statements, existence over number fields, and universal assertions about Hilbertian fields. The last distinction is essential in interpreting the counterexamples of Dittmann-Kadets. [132], [134], [121], [126], [129], [128], [131].

Recent results: boundaries worth recording

Fixed degree and height. Bhargava's published result settles the main van der Waerden counting problem in the monic coefficient box. State the O_n(H^(n-1)) exceptional bound for n ≥ 3; reducible monic quadratics instead have an H log H phenomenon. This regime has n fixed and H tending to infinity. [86], [81].

Growing degree and a fixed law. Breuillard-Varjú's main general theorem is conditional on RH for Dedekind zeta functions. The unconditional Bary-Soroker-Koukoulopoulos-Kozma results have specific hypotheses: for example, monic polynomials with independent coefficients uniform on a fixed interval of at least 35 integers are irreducible and have Galois group containing A_n with high probability, after excluding a zero constant term. Their broad no-small-factor theorem is not a universal high-probability irreducibility theorem. [91], [94].

Recent preprints. Kogler's version 2, dated 14 October 2026, states an unconditional proof for monic 0/1 polynomials with constant coefficient 1. Blachar-Breuillard-Kozma's version 1, dated 30 September 2026, proves a square-discriminant estimate under a high Rényi-entropy hypothesis and obtains full symmetric groups when combined with suitable A_n results. These are separate models and conclusions; neither statement should be silently replaced by a claim for every fixed coefficient law. [100], [99].

Reciprocal and matrix models. Reciprocal polynomials have a different ambient group, and matrix characteristic polynomials have dependent coefficients. Treat them as distinct branches. Retain the RH/ERH hypotheses in the cited random-matrix irreducibility papers. [79], [98], [103], [104].

Corrections and versions. The list includes Schinzel's sparse-polynomial erratum and the Ayad-McQuillan correction, and points to the correction of Dummit's quintic criterion. The reciprocal-polynomial counting paper's revised small-degree statement also matters. A theorem page in the atlas should link a correction beside the original result. [21], [124], [163], [79].

A useful structure for your research atlas

Give every method its own page: exact statement, assumptions, proof idea, a complete example, an example where it fails, links to related methods, and original references. Give every family a separate page for its defining formula and normalization, forced factors, known irreducibility results, known Galois groups, and open ranges.

Keep bibliographic records separate from mathematical claims. A source can support several theorems, and one theorem can have an original proof, a clearer exposition, and an erratum. For each claim record the ground field, degree range, parameter range, hypotheses, theorem number, proof method, publication status, and date last checked. Use the citation keys in the accompanying BibTeX and JSON files as stable identifiers.

For computations, store the exact polynomial, software and version, command, output, and certification status. For random models, also store the coefficient law, independence assumptions, conditioning, asymptotic variable, and quantitative error. A finite table of examples is evidence for exploration; a certified computation proves only the instances actually checked.

How to extend the bibliography

Search backward from Schinzel and the surveys; search forward from the exact DOI or title of each central paper. Use MathSciNet and zbMATH Open for bibliographic trails, arXiv for version histories, and publisher or author pages for final metadata. NUMDAM and EuDML are particularly useful for older European journals. Inspect corrections and the reference lists of newer papers before marking a theorem as current.

Useful MSC2020 headings include 12E05 (polynomials in general fields), 12E10 (special polynomials), 12E25 (Hilbertian fields and Hilbert irreducibility), 12F10 (Galois theory), and 12F12 (inverse Galois theory). Combine them with phrases such as Newton polygon, nonreciprocal part, discriminant, Frobenius cycle type, resolvent, prescribed coefficients, and arboreal representation.

The catalogue deliberately has no claim of exhaustiveness or automatic updating. In particular, it is not a complete bibliography of differential Galois theory, algebraic number field enumeration, or every recent variant of an irreducibility criterion. It is designed as a checked foundation that can grow by adding precise, linked records.

01 Foundations and reference books

[001] Local Fields

Jean-Pierre Serre. 1979. Springer, Graduate Texts in Mathematics 67.

Reference for valuations, ramification, the different, and discriminants. Supplies the local field framework behind Newton polygons and arguments that produce inertia elements in polynomial Galois groups.

Published book · Source | DOI 10.1007/978-1-4757-5673-9

[002] Finite Fields

Rudolf Lidl, Harald Niederreiter. 1997. Cambridge University Press, ed. 2, Encyclopedia of Mathematics and its Applications 20.

Standard reference for irreducible and primitive polynomials over finite fields, factorization, and constructions. Keep finite-field questions separate from Galois groups over the rationals: a degree-n irreducible polynomial over a finite field has cyclic splitting-field group of order n.

Published book · Source | DOI 10.1017/CBO9780511525926

The standard bibliographic year is 1997 (BNF and AMS review); the Cambridge product page lists a 24 October 1996 release date.

[003] Algebraic Number Theory

Jürgen Neukirch. 1999. Springer, Grundlehren der mathematischen Wissenschaften 322.

A source for prime decomposition, ramification, and global arithmetic of splitting fields. Useful when a polynomial argument passes from reductions modulo primes to arithmetic properties of number fields.

Published book · Source | DOI 10.1007/978-3-662-03983-0

[004] Polynomials with Special Regard to Reducibility

Andrzej Schinzel. 2000. Cambridge University Press, Encyclopedia of Mathematics and its Applications 77.

The central research monograph for this bibliography. It treats arbitrary and lacunary polynomials over several classes of fields; its bibliography is an essential starting point for backward searches into the classical literature.

Published book · Source | DOI 10.1017/CBO9780511542916

[005] Algebra

Serge Lang. 2002. Springer, ed. 3, Graduate Texts in Mathematics 211.

A reference for the algebra underlying irreducibility, field extensions, and Galois theory. Use it to check hypotheses over general fields and the distinction between separability and irreducibility.

Published book · Source | DOI 10.1007/978-1-4613-0041-0

[006] Polynomials

Victor V. Prasolov. 2004. Springer, Algorithms and Computation in Mathematics 11.

A broad treatment of polynomial techniques: irreducibility appears on pp. 47-76, special polynomial forms on pp. 77-131, and Galois theory on pp. 181-218. Useful for connecting algebraic arguments with root location and constructive examples.

Published book · Source | DOI 10.1007/978-3-642-03980-5

[007] Galois Theory

David A. Cox. 2012. John Wiley & Sons, ed. 2.

A detailed route from field extensions to explicit Galois groups. Chapters 13 and 14 treat resolvents, computation, and solvable permutation groups; especially useful for building worked examples.

Published book · Source | DOI 10.1002/9781118218457

[008] Fields and Galois Theory

James S. Milne. 2022. Kea Books.

A concise graduate reference with proofs and examples; Chapter 4 explains computation of Galois groups. The free course-note version 5.10 has the same mathematical content as the 2022 book with different pagination.

Published book · Source | Open copy

02 Classical criteria and sparse-polynomial methods

[009] Von denjenigen Moduln, welche Potenzen von Primzahlen sind. (Fortsetzung)

Theodor Schönemann. 1846. Journal für die reine und angewandte Mathematik 32, 93-105.

Historical source preceding Eisenstein for the criterion now often called Schönemann-Eisenstein. Read alongside Cox's historical account for the role of higher congruences and the precise translation into modern polynomial language.

published · Source | DOI 10.1515/crll.1846.32.93

[010] Über die Irreductibilität und einige andere Eigenschaften der Gleichung, von welcher die Theilung der ganzen Lemniscate abhängt

Gotthold Eisenstein. 1850. Journal für die reine und angewandte Mathematik 39, 160-179.

Original source for Eisenstein's treatment, motivated by division of the lemniscate and Gaussian integers. The familiar integer-polynomial criterion requires a prime dividing all nonleading coefficients, not the leading coefficient, and not dividing the constant coefficient twice.

published · Source | DOI 10.1515/crll.1850.39.160

[011] Sulla riduttibilità della funzione x^n-A in un campo qualunque di razionalità

Alfredo Capelli. 1901. Mathematische Annalen 54, 602-603.

Classical reference for the binomial irreducibility criterion, including the exceptional fourth-power obstruction when the exponent is divisible by four. Distinguish this binomial criterion from the related Capelli composition lemma; the note itself points back to Capelli's 1897 and 1898 work.

published · Source | DOI 10.1007/BF01450724

[012] Sur quelques cas d'irréductibilité des polynômes à coefficients rationnels

Gustave Dumas. 1906. Journal de Mathématiques Pures et Appliquées, 6 2, 191-258.

Foundational source for using Newton polygons to restrict possible factor degrees. In the familiar one-edge case, coprime horizontal and vertical increments give an irreducibility criterion; the polygon-product rule is useful far beyond this special case.

published · Source

[013] Neue Kriterien für die Irreduzibilität algebraischer Gleichungen

Oskar Perron. 1907. Journal für die reine und angewandte Mathematik 132, 288-307.

Classical criteria based on coefficient dominance and the location of complex roots. A standard application is a monic integer polynomial with nonzero constant term and next-to-leading coefficient larger in absolute value than one plus the sum of the lower coefficient magnitudes.

published · Source | DOI 10.1515/crll.1907.132.288

[014] Problem 226

Issai Schur. 1908. Archiv der Mathematik und Physik, 3 13, 367.

Original question on products of distinct integral linear factors perturbed by ±1. The link is to the later historical account that verifies this citation, rather than to a scan of the original problem; that account also records Schur's Problem275 (1909) on power compositions.

published; citation verified indirectly · Source

[015] Verschiedene Bemerkungen zur Zahlentheorie

George Pólya. 1919. Jahresbericht der Deutschen Mathematiker-Vereinigung 28, 31-40.

Original source for irreducibility criteria obtained from sufficiently small nonzero values at many distinct integers. The interpolation argument and later improvements are developed in Levit1968, Tverberg1973 and Győry-Hajdu-Tijdeman2011.

published · Source

[016] Zur Theorie der Irreduzibilitätskriterien

Øystein Ore. 1923. Mathematische Zeitschrift 18, 278-288.

An early systematic extension of valuation-based irreducibility criteria. This belongs in the chain from Eisenstein and Dumas to augmented valuations and higher-order Newton polygons.

published · Source

[017] Criteria for the irreducibility of polynomials

H. L. Dorwart, Øystein Ore. 1933. Annals of Mathematics 34(1), 81-94.

Studies irreducibility using repeated small or prime values at integer arguments, and extensions of Schur-type constructions. Particularly useful when the polynomial is naturally specified through its values or through products of linear factors.

published · Source | DOI 10.2307/1968341

[018] Irreducibility of polynomials

H. L. Dorwart. 1935. The American Mathematical Monthly 42(6), 369-381.

A compact historical survey organizing criteria through divisibility of coefficients, sizes of coefficients and values at integers. It remains a useful map of the classical literature, supplemented by later work for present-day coverage.

published · Source | DOI 10.1080/00029890.1935.11987732

[019] A construction for absolute values in polynomial rings

Saunders MacLane. 1936. Transactions of the American Mathematical Society 40(3), 363-395.

Constructs polynomial valuations through successive augmentations and key polynomials. It supplies a conceptual framework for irreducibility tests that go beyond a first Newton polygon, and underlies modern local factorization methods.

published · Source | DOI 10.1090/S0002-9947-1936-1501879-8

[020] On the reducibility of polynomials and in particular of trinomials

Andrzej Schinzel. 1965. Acta Arithmetica 11(1), 1-34.

A major classical source on structural reducibility questions, especially sparse polynomials and trinomials. Consult the 1966 errata together with the original paper before quoting a specific statement.

published; errata available · Source | DOI 10.4064/aa-11-1-1-34

[021] Errata to the paper "On the reducibility of polynomials and in particular of trinomials"

Andrzej Schinzel. 1966. Acta Arithmetica 11(4), 491.

Corrections to Schinzel's 1965 paper. Keep this linked to the original record in any atlas so readers do not use an uncorrected statement.

published erratum · Source | DOI 10.4064/aa-11-4-491

[022] Irreducibility of polynomials with low absolute values

R. J. Levit. 1968. Transactions of the American Mathematical Society 132(2), 297-305.

Sharpens the interpolation bounds behind Pólya's small-value criterion. It also excludes factors of specified degrees, which is often more useful than an all-or-nothing test when combined with local information.

published · Source | DOI 10.2307/1994841

[023] On the irreducibility of polynomials taking small values

Helge Tverberg. 1973. Mathematica Scandinavica 32, 5-21.

Further development of the small-value method, including asymptotic refinements. Read after Pólya and Levit when integer evaluations give stronger information than coefficient divisibility.

published · Source | DOI 10.7146/math.scand.a-11442

[024] On an irreducibility theorem of A. Cohn

John Brillhart, Michael Filaseta, Andrew Odlyzko. 1981. Canadian Journal of Mathematics 33(5), 1055-1059.

Generalizes the criterion that a prime's digit expansion yields an irreducible polynomial when the digits are used as coefficients. Useful for prime-value methods and for understanding precisely which coefficient restrictions make a single prime evaluation decisive.

published · Source | DOI 10.4153/CJM-1981-080-0

[025] On a generalization of Eisenstein's irreducibility criterion

Sudesh K. Khanduja, Jayanti Saha. 1997. Mathematika 44(1), 37-41.

A modern extension of the Eisenstein method in the setting of valued fields. It is a useful entry to generalizations where the ordinary divisibility pattern of the coefficients does not suffice.

published · Source | DOI 10.1112/S0025579300011931

[026] On the factorization of polynomials with small Euclidean norm

Michael Filaseta. 1999. Number Theory in Progress, Vol. 1, 143-163.

Uses restrictions on the Euclidean norm of the coefficient vector to control factorization. A useful methodological source for sparse polynomials and for results about their nonreciprocal parts, which must be distinguished from irreducibility of the whole polynomial.

published · Source | DOI 10.1515/9783110285581.143

Pages143-163 follow the author's bibliography and MathSciNet citation reproduced in Sawin-Shusterman-Stoll; the publisher chapter wrapper gives143-164, apparently including a trailing page.

[027] On an irreducibility theorem of A. Schinzel associated with coverings of the integers

Michael Filaseta, Kevin Ford, Sergei Konyagin. 2000. Illinois Journal of Mathematics 44(3), 633-643.

For large exponents, controls the nonreciprocal part of f(x)x^n+g(x), assuming coprimality, nonzero constant terms of both f and g, and explicit Capelli-type exclusions. The method improves the threshold in Schinzel's theorem and connects sparse factorization to covering systems.

published · Source | DOI 10.1215/ijm/1256060421

[028] Prime numbers and irreducible polynomials

M. Ram Murty. 2002. The American Mathematical Monthly 109(5), 452-458.

Readable account linking prime values, coefficient bounds and irreducibility, with Cohn's theorem as a central example. A single prime value needs additional size or coefficient hypotheses; the paper provides such hypotheses rather than asserting a general converse to prime-producing conjectures.

published · Source | DOI 10.1080/00029890.2002.11919872

[029] Irreducibility of polynomials whose coefficients are integers

R. Thangadurai. 2007. RMS Mathematical Newsletter 17(2), 29-37.

Accessible survey of rational irreducibility for integer polynomials, covering several classical tests and their limitations. A useful first reading before the more specialized Newton-polygon, small-value and prime-value papers in this section.

published · Source

[030] Why Eisenstein Proved the Eisenstein Criterion and Why Schönemann Discovered It First

David A. Cox. 2011. The American Mathematical Monthly 118(1), 3-21.

A particularly useful historical and mathematical exposition linking Gauss's lemma, finite fields, Hensel lifting, Gaussian integers and lemniscatic division. It gives careful attribution and explains the distinct routes by which Schönemann and Eisenstein arrived at their criteria.

published · Source | DOI 10.4169/amer.math.monthly.118.01.003

[031] Irreducibility criteria of Schur-type and Pólya-type

Kálmán Győry, Lajos Hajdu, Robert Tijdeman. 2011. Monatshefte für Mathematik 163(4), 415-443.

Research paper with a substantial historical introduction and bibliography on small-value criteria, perturbed products and compositions. Its extensions use interpolation, algebraic number theory and Diophantine approximation, including results over imaginary quadratic rings and compositions of CM type.

published · Source | DOI 10.1007/s00605-010-0241-9

[032] Schönemann-Eisenstein-Dumas-type irreducibility conditions that use arbitrarily many prime numbers

Nicolae Ciprian Bonciocat. 2015. Communications in Algebra 43(8), 3102-3122.

Combines coefficient divisibility information from several primes through Newton polygons. This is directly useful when each prime leaves possible factor degrees but their combined restrictions exclude every proper factor.

published · Source | DOI 10.1080/00927872.2014.910800

[033] Further irreducibility criteria for polynomials with non-negative coefficients

Morgan Cole, Scott Dunn, Michael Filaseta. 2016. Acta Arithmetica 175(2), 137-181.

Continues the Cohn prime-value approach for nonnegative integer coefficients. It determines sharp coefficient bounds for bases from 3 through 20, allowing coefficients larger than ordinary base digits while retaining an irreducibility conclusion from a prime evaluation.

published · Source | DOI 10.4064/aa8376-5-2016

[034] On the irreducibility of the non-reciprocal part of polynomials of the form f(x)x^n+g(x)

Michael Filaseta, Huixi Li, Frank Patane, Dane Skabelund. 2020. Acta Arithmetica 196(2), 187-201.

Improves the sufficient exponent threshold in the robust-pair method of Sawin, Shusterman and Stoll. The conclusion concerns the nonreciprocal part, so reciprocal factors still require separate analysis when proving full irreducibility.

published · Source | DOI 10.4064/aa190907-11-2

[035] Irreducibility of polynomials with a large gap

Will Sawin, Mark Shusterman, Michael Stoll. 2020. Acta Arithmetica 192(2), 111-139.

Develops Ljunggren's factor-reversal method into practical irreducibility algorithms for x^N c(x^-1)+d(x). It treats cyclotomic obstructions separately and improves effective bounds under additional robustness hypotheses; its introduction is a useful guide to earlier large-gap work.

published · Source | DOI 10.4064/aa180526-12-6 | arXiv:1803.10811

03 Galois group detection and effective prime methods

Permutation groups and Frobenius methods

[036] Théorèmes sur les groupes primitifs

Camille Jordan. 1871. Journal de Mathématiques Pures et Appliquées (2) 16, 383-408.

Historical source for the primitive-group results underlying many proofs that a polynomial's Galois group contains A_n. Use modern expositions for the precise prime-cycle criterion and for checking primitivity before applying it.

Published article · Source

[037] Permutation Groups

John D. Dixon, Brian Mortimer. 1996. Graduate Texts in Mathematics 163.

Reference for transitivity, block systems, primitive groups, and the Jordan-type results used to force large Galois groups. It supports the group-theoretic step after reduction modulo primes or local ramification has supplied cycle information.

Published monograph · Source | DOI 10.1007/978-1-4612-0731-3

[038] Chebotarëv and his density theorem

Peter Stevenhagen, Jr., Hendrik W. Lenstra. 1996. The Mathematical Intelligencer 18(2), 26-37.

An accessible explanation of how splitting modulo primes records conjugacy classes in a Galois group, together with the history and proof of Chebotarev's theorem. A strong source for explaining both the power and the limitations of modular factorization experiments.

Published expository article · Source | DOI 10.1007/BF03027290

[039] Primitive permutation groups containing a cycle

Gareth A. Jones. 2014. Bulletin of the Australian Mathematical Society 89(1), 159-165.

Classifies finite primitive permutation groups containing a cycle and gives modern refinements of Jordan's theorem. Useful when a modular factorization supplies a cycle whose length is not prime; primitivity remains an essential hypothesis.

Published article · Source | DOI 10.1017/S000497271300049X | arXiv:1209.5169

Effective Chebotarev and quantitative specialization

[040] Effective versions of the Chebotarev density theorem

Jeffrey C. Lagarias, Andrew M. Odlyzko. 1977. Algebraic Number Fields: L-functions and Galois Properties (Proc. Sympos., Univ. Durham, Durham, 1975), 409-464.

A foundational source for effective forms of Chebotarev, with conditional and unconditional estimates distinguished. It explains the analytic input needed to turn existence of useful Frobenius elements into bounds on the primes where they occur.

Published proceedings chapter · Source

[041] The Distribution of Galois Groups and Hilbert's Irreducibility Theorem

Stephen D. Cohen. 1981. Proceedings of the London Mathematical Society (3) 43(2), 227-250.

A foundational quantitative treatment of exceptional specializations and the distribution of Galois groups in polynomial families. It connects Hilbert irreducibility with sieve estimates, complementing the qualitative existence theorem.

Published article · Source | DOI 10.1112/plms/s3-43.2.227

[042] Quelques applications du théorème de densité de Chebotarev

Jean-Pierre Serre. 1981. Publications Mathématiques de l'IHÉS 54, 123-201.

Develops effective Chebotarev estimates and arithmetic applications. An important reference when checking exactly how analytic hypotheses, extension degrees, and discriminants enter arguments based on observed factorization types modulo primes.

Published article · Source | DOI 10.1007/BF02698692

[043] A unified and improved Chebotarev density theorem

Jesse Thorner, Asif Zaman. 2019. Algebra & Number Theory 13(5), 1039-1068.

Gives an unconditional effective Chebotarev theorem with improved uniformity, including treatment of a possible exceptional zero. Useful for evaluating whether a probabilistic or computational Galois argument can replace a GRH input by an unconditional estimate within its required parameter range.

Published article · Source | DOI 10.2140/ant.2019.13.1039 | arXiv:1803.02823

04 Special polynomial families

trinomials and sparse families

[044] On the irreducibility of certain trinomials

Ernst S. Selmer. 1956. Mathematica Scandinavica 4, 287-302.

The classical source for irreducibility of x^n-x-1 over Q for every n at least 2. A starting point for the arithmetic of sparse polynomials; the full symmetric-group conclusion has a separate attribution.

Published · Source | DOI 10.7146/math.scand.a-10478

[045] On the irreducibility of certain trinomials and quadrinomials

Wilhelm Ljunggren. 1960. Mathematica Scandinavica 8, 65-70.

A foundational source for factorization of sparse polynomials and the reciprocal/nonreciprocal distinction. The factor-reversal method developed here has later applications far beyond these initial families.

Published · Source | DOI 10.7146/math.scand.a-10593

[046] Equations of the type X^n+aX+b with absolute Galois group S_n

Enric Nart, Núria Vila. 1979. Revista de la Universidad de Santander 2, 821-825.

An early source for symmetric Galois groups of trinomials; in particular, it precedes Osada for x^n-x-1. The institutional catalogue supplies an English title although the original paper is in Spanish.

Published · Source

[047] The Galois groups of the polynomials X^n+aX^l+b

Hiroyuki Osada. 1987. Journal of Number Theory 25(2), 230-238.

Gives arithmetic conditions forcing the full symmetric group for trinomials. Useful for the ramification route to transpositions and for the independent proof concerning x^n-x-1.

Published · Source | DOI 10.1016/0022-314X(87)90029-1

[048] Double transitivity of Galois groups of trinomials

Stephen D. Cohen, A. Movahhedi, Alain Salinier. 1997. Acta Arithmetica 82(1), 1-15.

Studies how arithmetic hypotheses on a trinomial force double transitivity. This is an intermediate group-theoretic step that can be combined with cycle or ramification information to obtain a large Galois group.

Published · Source | DOI 10.4064/aa-82-1-1-15

[049] Galois groups of trinomials

Stephen D. Cohen, A. Movahhedi, Alain Salinier. 1999. Journal of Algebra 222(2), 561-573.

Extends earlier results of Osada using primitivity, cycle structure, and arithmetic restrictions. Apply the precise irreducibility and coprimality assumptions in the paper rather than treating sparsity alone as a sufficient condition.

Published · Source | DOI 10.1006/jabr.1999.8033

Schur and orthogonal polynomials

[050] Einige Sätze über Primzahlen mit Anwendungen auf Irreduzibilitätsfragen, I

Issai Schur. 1929. Sitzungsberichte der Preußischen Akademie der Wissenschaften, Physikalisch-Mathematische Klasse, 125-136.

Classical prime-divisor arguments applied to irreducibility. Read alongside Part II and the 1930/1931 papers to trace the origin of factorial-coefficient and orthogonal-polynomial methods.

Published · Source

[051] Gleichungen ohne Affekt

Issai Schur. 1930. Sitzungsberichte der Preußischen Akademie der Wissenschaften, Physikalisch-Mathematische Klasse, 443-449.

Classical source for the Galois-group theorem for exponential Taylor polynomials. Coleman gives a particularly accessible later proof through Newton polygons.

Published · Source

[052] Affektlose Gleichungen in der Theorie der Laguerreschen und Hermiteschen Polynome

Issai Schur. 1931. Journal für die reine und angewandte Mathematik 165, 52-58.

Foundational treatment of irreducibility and Galois groups in the Laguerre and Hermite families. For Hermite polynomials, parity and the forced factor at zero must be respected in any statement about irreducibility.

Published · Source | DOI 10.1515/crll.1931.165.52

Taylor and truncated binomial polynomials

[053] On the Galois groups of the exponential Taylor polynomials

Robert F. Coleman. 1987. L'Enseignement Mathématique 33, 183-189.

A short Newton-polygon proof of Schur’s theorem for sum_{j=0}^n x^j/j!: irreducible over Q, with group A_n when 4 divides n and S_n otherwise (n at least 2). A model example of local slopes, long cycles, and the discriminant square test working together.

Published · Source | DOI 10.5169/seals-87891

[054] The Galois group of x^n-x^{n-1}-...-x-1

Paulo A. Martin. 2004. Journal of Pure and Applied Algebra 190(1--3), 213-223.

Proves that the generalized Fibonacci polynomial has group S_n when n is even or prime. The prime-degree argument includes irreducibility of its reduction modulo that prime.

Published · Source | DOI 10.1016/j.jpaa.2003.10.028

[055] On the irreducibility of a truncated binomial expansion

Michael Filaseta, Angel Kumchev, Dmitrii Pasechnik. 2007. Rocky Mountain Journal of Mathematics 37(2), 455-464.

Investigates the first k+1 terms of (1+x)^n and related coefficient families. A useful bridge from factorial-coefficient criteria to explicit truncated-binomial questions.

Published · Source | DOI 10.1216/rmjm/1181068761

[056] The theorem of Jentzsch-Szegő on an analytic curve: application to the irreducibility of truncations of power series

Antoine Chambert-Loir. 2011. International Journal of Number Theory 7(7), 1807-1823.

Relates zero-distribution on analytic curves, including nonarchimedean curves, to irreducibility information for truncations. Use the corrected published version or arXiv v3; the arXiv history specifically records a correction to the application.

Published · Source | DOI 10.1142/S1793042111004691 | arXiv:1004.3813

[057] On the Galois group over Q of a truncated binomial expansion

Michael Filaseta, Richard Moy. 2018. Colloquium Mathematicum 154(2), 295-308.

For a fixed truncation degree r other than 6, proves that sufficiently large n gives S_r for the first r+1 terms of (1+x)^n. At r=6 it bounds the number of exceptions n up to N by O(log N).

Published · Source | DOI 10.4064/cm7474-3-2018

[058] Galois groups of Taylor polynomials of some elementary functions

Khosro Monsef Shokri, Jafar Shaffaf, Reza Taleb. 2019. International Journal of Number Theory 15(6), 1127-1141.

Studies truncations associated with logarithmic and cosine functions. The cosine examples naturally involve a signed-permutation group because of their even symmetry.

Published · Source | DOI 10.1142/S1793042119500623

[059] Exponentielle tronquée et autres contes galoisiens

Patrick Rabarison, Fabien Pazuki, Pascal Molin. 2024. Publications mathématiques de Besançon. Algèbre et théorie des nombres, 105-117.

A compact survey centred on exponential truncations, with further explicit investigations of Padé approximants. Particularly useful as an entry point connecting classical Schur results with later computational questions.

Published · Source | DOI 10.5802/pmb.57

[060] Speculations in Galois Theory

Nicholas M. Katz, Igor Rivin. undated. Author-hosted note.

Poses limiting-frequency questions for Galois groups of Taylor truncations and reports computations in many families. The note explicitly separates experimental evidence from theorems; its posted copy has no date on the title page.

Author-hosted contribution to the Laumon volume; publication year not verified · Source

Laguerre and Bessel polynomials

[061] Bessel Polynomials

Emil Grosswald. 1978. Springer, Lecture Notes in Mathematics 698.

The standard monograph, including chapters on algebraic irreducibility and the Galois group. Read its conjectural irreducibility discussion together with the later Filaseta-Trifonov theorem.

Published · Source | DOI 10.1007/BFb0063135

[062] On the irreducibility of the generalized Laguerre polynomials

Michael Filaseta, Tsit-Yuen Lam. 2002. Acta Arithmetica 105(2), 177-182.

For each fixed rational parameter outside the negative integers, proves irreducibility over Q in every sufficiently large degree. The threshold depends on the parameter, so this is not a uniform theorem when the parameter varies with degree.

Published · Source | DOI 10.4064/aa105-2-4

[063] The irreducibility of the Bessel polynomials

Michael Filaseta, Ognian Trifonov. 2002. Journal für die reine und angewandte Mathematik 550, 125-140.

Resolves Grosswald’s irreducibility conjecture for Bessel polynomials in every positive degree. Proves a stronger coefficient-perturbation statement using Newton polygons and prime-divisor estimates.

Published · Source | DOI 10.1515/crll.2002.069

[064] On the Galois group of generalized Laguerre polynomials

Farshid Hajir. 2005. Journal de théorie des nombres de Bordeaux 17(2), 517-525.

Introduces a Newton-index criterion for containing A_n and applies it, together with irreducibility results, to generalized Laguerre polynomials with fixed rational parameter outside the negative integers in sufficiently large degree. Irreducibility is an explicit input to the group criterion.

Published · Source | DOI 10.5802/jtnb.505 | arXiv:math/0406308

[065] Algebraic properties of a family of generalized Laguerre polynomials

Farshid Hajir. 2009. Canadian Journal of Mathematics 61(3), 583-603.

Studies negative integral parameters linked to degree, a regime different from the fixed-parameter theorem. Newton polygons organize irreducibility and large-group results, together with precise remaining conjectures.

Published · Source | DOI 10.4153/CJM-2009-031-6 | arXiv:math/0406307

[066] Laguerre polynomials with Galois group A_m for each m

Michael Filaseta, Travis Kidd, Ognian Trifonov. 2012. Journal of Number Theory 132, 776-805.

Completes Schur's generalized Laguerre realizations of A_m by treating m congruent to 2 modulo 4. The constructed polynomials are irreducible for $m > 2$, while $m = 2$ is the split case.

Published · Source | DOI 10.1016/j.jnt.2011.09.012

[067] On classifying Laguerre polynomials which have Galois group the alternating group

Pradipto Banerjee, Michael Filaseta, Carrie E. Finch, J. Russell Leidy. 2013. Journal de théorie des nombres de Bordeaux 25(1), 1-30.

Classifies the integer parameter pairs with nonzero square discriminant into explicit infinite families and a finite remainder. A square discriminant constrains the group to A_n but does not by itself prove irreducibility or equality with A_n.

Published · Source | DOI 10.5802/jtnb.822

[068] Families of Laguerre polynomials with alternating group as Galois group

Ankita Jindal, Shanta Laishram. 2022. Journal of Number Theory 241, 387-429.

Proves irreducibility and group results for integer parameters in the interval [-2n-4,-2n], including Bessel-related cases. Verifies parts of the alternating-group conjectural classification in that range.

Published · Source | DOI 10.1016/j.jnt.2022.04.001

Jacobi, Legendre and related polynomials

[069] Algebraic properties of a family of Jacobi polynomials

John Cullinan, Farshid Hajir, Elizabeth Sell. 2009. Journal de théorie des nombres de Bordeaux 21(1), 97-108.

For fixed n at least 6, obtains irreducibility for all but finitely many rational parameters in a particular one-parameter Jacobi subfamily. The Galois-group exceptional set is finite for odd n and thin for even n.

Published · Source | DOI 10.5802/jtnb.659

[070] On the Galois groups of Legendre polynomials

John Cullinan, Farshid Hajir. 2014. Indagationes Mathematicae 25(3), 534-552.

Studies the groups of parity-reduced Legendre polynomials through discriminants and Newton polygons. Its results and conjectures must be stated for the correct reduced polynomial, with irreducibility assumptions where required.

Published · Source | DOI 10.1016/j.indag.2014.01.004

[071] Algebraic properties of Kaneko-Zagier lifts of supersingular polynomials

John Cullinan, Farshid Hajir. 2017. Proceedings of the American Mathematical Society 145(6), 2291-2304.

Connects rational lifts of supersingular polynomials with specialized Jacobi polynomials. Gives new irreducibility cases and investigates the Mahlburg-Ono maximal-Galois-group conjecture after removing trivial factors.

Published · Source | DOI 10.1090/proc/13212

[072] On the irreducibility of the Krawtchouck polynomials

John Cullinan. 2024. Journal of the Ramanujan Mathematical Society 39(4), 323-335.

Determines Newton polygons for certain Krawtchouck polynomials and proves new irreducibility cases. Shows how a familiar local method applies to a less understood discrete orthogonal family.

Published · Source | arXiv:2212.07539

Euler, Bernoulli and Eulerian polynomials

[073] Note on irreducibility of the Bernoulli and Euler polynomials

Leonard Carlitz. 1952. Duke Mathematical Journal 19, 475-481.

An early arithmetic source for irreducibility in Bernoulli and Euler families. These are the Appell polynomials defined by exponential generating functions, and should not be conflated with descent-enumerating Eulerian polynomials.

Published · Source | DOI 10.1215/S0012-7094-52-01948-0

[074] On the Euler and Bernoulli polynomials

John Brillhart. 1969. Journal für die reine und angewandte Mathematik 234, 45-64.

A central classical paper on arithmetic factorization and roots of Euler and Bernoulli polynomials. Useful both for known irreducibility results and for the normalization needed after removing forced factors.

Published · Source | DOI 10.1515/crll.1969.234.45

[075] Théorie géométrique des polynômes eulériens

Dominique Foata, Marcel-Paul Schützenberger. 1970. Springer, Lecture Notes in Mathematics 138.

Foundational combinatorial background for Eulerian polynomials: conventions, generating functions, and symmetry. It supplies the structural background needed before formulating an arithmetic irreducibility or Galois question.

Published · Source

[076] On the irreducibility of a class of Euler Frobenius polynomials

A. Sharma, Ernst G. Straus. 1974. Canadian Mathematical Bulletin 17(2), 265-273.

A key reference for Euler-Frobenius irreducibility. Theorem 6 at parameter r=1 includes the classical Eulerian polynomial at an odd prime index, where a shift makes Eisenstein’s criterion visible.

Published · Source | DOI 10.4153/CMB-1974-052-x

Fekete and reciprocal polynomials

[077] Fekete polynomials, quadratic residues, and arithmetic

Ján Mináč, Tung T. Nguyen, Nguyen Duy Tân. 2023. Journal of Number Theory 242, 669-708.

Introduces normalized reciprocal and trace polynomials associated with prime-conductor Fekete polynomials, relates values to class numbers, and studies their splitting fields. The proposed full wreath-product group is a conjecture in this source, not an all-prime theorem.

Published · Source | DOI 10.1016/j.jnt.2022.05.001 | arXiv:2111.05256

[078] On the arithmetic of generalized Fekete polynomials

Ján Mináč, Tung T. Nguyen, Nguyen Duy Tân. 2024. Experimental Mathematics 33(4), 723-754.

Extends the arithmetic investigation to quadratic characters with possibly composite conductor. Cyclotomic factors, trace-polynomial reductions, and extra symmetries constrain the possible Galois groups.

Published · Source | DOI 10.1080/10586458.2023.2293283 | arXiv:2206.11778

[079] Galois groups of reciprocal polynomials and the van der Waerden-Bhargava theorem

Theresa C. Anderson, Adam Bertelli, Evan M. O'Dorney. 2026. Algebra & Number Theory 20, 603-628.

For separable reciprocal polynomials of degree 2n and growing coefficient height H, gives exceptional proportion of order H^(-1) log H for nonmonic n ≥ 1 and monic n ≥ 3. The corrected version leaves the monic quartic case open; the expected group is the full signed-permutation group.

Published · Source | DOI 10.2140/ant.2026.20.603 | arXiv:2406.18970

05 Counting, random polynomials, and random matrices

Counting by height and probabilistic Galois theory

[080] Die Seltenheit der reduziblen Gleichungen und der Gleichungen mit Affekt

B. L. van der Waerden. 1936. Monatshefte für Mathematik und Physik 43, 133-147.

Classical starting point for the rarity of reducible polynomials and exceptional Galois groups in coefficient boxes. The asymptotic variable is coefficient height, with degree fixed.

Published · Source | DOI 10.1007/BF01707594

[081] Reducible polynomials

R. Chela. 1963. Journal of the London Mathematical Society 38(1), 183-188.

Classical asymptotic counting of reducible integer polynomials in fixed-degree coefficient boxes. Useful for the main contribution from linear factors and for separating reducibility from exceptional transitive Galois groups.

Published · Source | DOI 10.1112/jlms/s1-38.1.183

[082] The large sieve and probabilistic Galois theory

P. X. Gallagher. 1973. Analytic Number Theory, Proceedings of Symposia in Pure Mathematics 24, 91-101.

Foundational use of the large sieve to combine factorization information at many primes and bound exceptional Galois groups. The method turns finite-field factorization statistics into quantitative results for fixed-degree integer coefficient boxes.

Published · Source | DOI 10.1090/pspum/024/0332694

[083] On the distribution of Galois groups

Rainer Dietmann. 2012. Mathematika 58(1), 35-44.

Bounds the number of monic integer polynomials of fixed degree and bounded height with a prescribed Galois group. Useful for the resolvent and integral-point approach to exceptional groups.

Published · Source | DOI 10.1112/S0025579311002105 | arXiv:1010.5341

[084] Enumerative Galois theory for cubics and quartics

Sam Chow, Rainer Dietmann. 2020. Advances in Mathematics 372, 107282.

Studies counts for individual Galois groups in degrees three and four. A useful source for understanding the stronger asymptotic questions that remain after a generic full-symmetric-group result.

Published · Source | DOI 10.1016/j.aim.2020.107282 | arXiv:1807.05820

[085] Probabilistic Galois Theory: The square discriminant case

Lior Bary-Soroker, Or Ben-Porath, Vlad Matei. 2024. Bulletin of the London Mathematical Society 56(6), 2162-2177.

Studies the frequency of alternating Galois groups and nonzero square discriminants in the fixed-degree large-box model. Constructions and lower bounds show why a naive random-integer model for the discriminant can be misleading.

Published · Source | DOI 10.1112/blms.13049 | arXiv:2207.12493

[086] Galois groups of random integer polynomials and van der Waerden's Conjecture

Manjul Bhargava. 2025. Annals of Mathematics 201(2), 339-377.

Proves the order-of-magnitude bound O(H^(n-1)) for monic integer polynomials of each fixed degree $n \geq 3$ and height at most H whose Galois group is not S_n. Degree is fixed; this does not settle the fixed-coefficient-law, growing-degree problem.

Published · Source | DOI 10.4007/annals.2025.201.2.1 | arXiv:2111.06507

Random polynomials of growing degree

[087] Zeros of polynomials with 0,1 coefficients

A. M. Odlyzko, Bjorn Poonen. 1993. L'Enseignement Mathématique 39(3--4), 317-348.

Historical source for the asymptotic irreducibility question for monic 0-1 polynomials with constant coefficient 1. Its principal results concern the geometry and topology of their zero sets, which provide useful background for restricted-coefficient families.

Published · Source

[088] On the number of irreducible polynomials with 0,1 coefficients

S. V. Konyagin. 1999. Acta Arithmetica 88(4), 333-350.

A foundational unconditional result for the 0-1 model, excluding small-degree factors with high probability and obtaining a lower bound of order 1/log n for irreducibility. The finite-field and exponential-sum techniques are important predecessors of later work.

Published · Source | DOI 10.4064/aa-88-4-333-350

[089] Is a bivariate polynomial with ±1 coefficients irreducible? Very likely!

Lior Bary-Soroker, Gady Kozma. 2017. International Journal of Number Theory 13(4), 933-936.

A short treatment of irreducibility for random bivariate sign polynomials. It is useful for comparing the extra flexibility of several variables with the substantially harder univariate fixed-coefficient problem.

Published · Source | DOI 10.1142/S1793042117500488

[090] Irreducibility of random polynomials

Christian Borst, Evan Boyd, Claire Brekken, Samantha Solberg, Melanie Matchett Wood, Philip Matchett Wood. 2018. Experimental Mathematics 27(4), 498-506.

Computational study of several sampling models and a universality heuristic relating reducibility to low-degree factors. Treat its numerical evidence and conjectures separately from its rigorous statements.

Published · Source | DOI 10.1080/10586458.2017.1325790 | arXiv:1705.03709

[091] Irreducibility of random polynomials of large degree

Emmanuel Breuillard, Péter P. Varjú. 2019. Acta Mathematica 223(2), 195-249.

Under Riemann hypotheses for suitable Dedekind zeta functions, proves irreducibility and alternating-group containment for broad independent-coefficient models, with careful treatment of zero and cyclotomic factors. The method connects finite-field random walks, root counts and effective prime-ideal estimates.

Published; main general results conditional on RH · Source | DOI 10.4310/ACTA.2019.v223.n2.a1 | arXiv:1810.13360

[092] Low-degree factors of random polynomials

Sean O'Rourke, Philip Matchett Wood. 2019. Journal of Theoretical Probability 32(2), 1076-1104.

Converts pointwise delocalization of roots into exclusion of factors of low algebraic degree. Covers several random-matrix characteristic-polynomial models as well as random polynomials; absence of low-degree factors alone is weaker than irreducibility.

Published · Source | DOI 10.1007/s10959-018-0839-8 | arXiv:1608.01938

[093] Irreducible polynomials of bounded height

Lior Bary-Soroker, Gady Kozma. 2020. Duke Mathematical Journal 169(4), 579-598.

Proves high-probability irreducibility and containment of A_n for suitable fixed coefficient distributions, including uniform coefficients in {1,...,210}. Combines reductions at several primes with probabilistic permutation-group arguments.

Published · Source | DOI 10.1215/00127094-2019-0047 | arXiv:1710.05165

[094] Irreducibility of random polynomials: general measures

Lior Bary-Soroker, Dimitris Koukoulopoulos, Gady Kozma. 2023. Inventiones Mathematicae 233(3), 1041-1120.

Develops Fourier equidistribution, a sieve for factor degrees and permutation-group methods for independent integer coefficients, conditioning on nonzero constant term. Proves high-probability irreducibility and Galois group A_n or S_n under explicit distributional hypotheses, including a uniform interval of at least 35 integers; much more general laws yield exclusion of factors up to a positive fraction of the degree.

Published · Source | DOI 10.1007/s00222-023-01193-6 | arXiv:2007.14567

[095] Full Galois groups of polynomials with slowly growing coefficients

Lior Bary-Soroker, Noam Goldgraber. 2025. Bulletin of the London Mathematical Society 57(3), 941-955.

For monic degree-n polynomials with coefficients uniform in [-L(n),L(n)], proves full symmetric Galois group with high probability whenever L(n) tends to infinity. The interval may grow arbitrarily slowly; the theorem does not assert the same conclusion for a fixed interval.

Published · Source | DOI 10.1112/blms.70008 | arXiv:2404.13559

[096] Irreducibility of Littlewood polynomials of special degrees

Lior Bary-Soroker, David Hokken, Gady Kozma, Bjorn Poonen. 2025. International Mathematics Research Notices 2025(21), rnaf326.

Proves unconditionally that the limsup of the irreducibility probability for uniformly random Littlewood polynomials is 1. Special-degree factorization constraints are essential; the statement should not be replaced by an all-degree limit.

Published · Source | DOI 10.1093/imrn/rnaf326 | arXiv:2308.04878

[097] Irreducibility of random polynomials of Z[X]

Pierre-Alexandre Bazin. 2025. International Mathematics Research Notices 2025(11), rnaf136.

Improves rates in the Bary-Soroker-Koukoulopoulos-Kozma framework when near-uniformity holds modulo more primes. The zero-constant-term obstruction remains: for example, uniform intervals of length at least 10^8 give P(reducible) ≤ P(a_0=0)+O(n^(-1/2)), rather than an unconditioned bound tending to zero.

Published · Source | DOI 10.1093/imrn/rnaf136 | arXiv:2402.06643

[098] Irreducibility and Galois groups of random reciprocal polynomials of large degree

David Hokken, Dimitris Koukoulopoulos. 2025.

For reciprocal polynomials of degree 2m and coefficient laws satisfying a Fourier hypothesis, proves high-probability irreducibility and a Galois group equal to the hyperoctahedral group or one of two index-two subgroups. The paired-root symmetry changes both the natural ambient group and the factorization arguments.

Preprint; version 2 dated 23 September 2026 · Source | arXiv:2510.18857

[099] Galois groups of random polynomials of large degree

Guy Blachar, Emmanuel Breuillard, Gady Kozma. 2026.

Preprint proving an O(1/log n) bound for square-discriminant probability when the bounded integer coefficient law has Rényi entropy of order 2 greater than 12. Its corollary gives full S_n with high probability for coefficients uniform on a fixed interval of N > exp(12) integers, conditioned on nonzero constant term; the fair-sign law is not covered.

Preprint; version 1 dated 30 September 2026 · Source | arXiv:2609.40123

[100] The Odlyzko-Poonen Conjecture on Irreducibility of Random Polynomials

Constantin Kogler. 2026.

Preprint presenting an unconditional proof of the irreducibility limit for monic polynomials with constant coefficient 1 and all remaining coefficients independent uniform in {0,1}. This precise model and the preprint status should accompany any account of the claimed advance.

Preprint; version 2 dated 14 October 2026 · Source | arXiv:2609.26771

[101] A random polynomial with multiplicative coefficients is almost surely irreducible

Péter P. Varjú, Max Wenqiang Xu. 2026. International Mathematics Research Notices 2026(16), rnag178.

Under RH for Dedekind zeta functions, proves irreducibility with probability 1-O(d^(-1/2+epsilon)) for specified random multiplicative-coefficient models. The theorem concerns P_d(x)/x, after removal of the obligatory factor x; these coefficients are dependent.

Published; conditional on RH · Source | DOI 10.1093/imrn/rnag178 | arXiv:2511.04240

Characteristic polynomials and random matrices

[102] Walks on groups, counting reducible matrices, polynomials, and surface and free group automorphisms

Igor Rivin. 2008. Duke Mathematical Journal 142(2), 353-379.

Relates random walks on arithmetic groups and finite quotients to irreducible characteristic polynomials. Provides a route from arithmetic of polynomials to geometric group theory; its random-walk and height models differ from independent bounded matrix entries.

Published · Source | DOI 10.1215/00127094-2008-009 | arXiv:math/0703532

[103] The characteristic polynomial of a random matrix

Sean Eberhard. 2022. Combinatorica 42(4), 491-527.

Assuming ERH, proves high-probability irreducibility and containment of A_n for characteristic polynomials of matrices with independent entries from a fixed nontrivial finitely supported integer law. The dependent polynomial coefficients require matrix-specific input.

Published; conditional on ERH · Source | DOI 10.1007/s00493-020-4657-0 | arXiv:2008.01223

[104] Random symmetric matrices: rank distribution and irreducibility of the characteristic polynomial

Asaf Ferber, Vishesh Jain, Ashwin Sah, Mehtaab Sawhney. 2023. Mathematical Proceedings of the Cambridge Philosophical Society 174(2), 233-246.

Under ERH, proves high-probability irreducibility for characteristic polynomials of random symmetric sign matrices. The key innovation is sharp rank-distribution estimates over finite fields, using inverse Littlewood-Offord ideas.

Published; irreducibility application conditional on ERH · Source | DOI 10.1017/S0305004122000226 | arXiv:2106.04049

[105] Galois groups of random integer matrices

Theresa C. Anderson, Evan M. O'Dorney. 2026. Mathematical Proceedings of the Cambridge Philosophical Society, 1-16.

Studies fixed-size integer matrices with entries bounded by a height parameter and counts characteristic polynomials with non-symmetric Galois group. Uses the large sieve, Fourier analysis and geometric sieving; this is a different asymptotic regime from Eberhard's growing-dimension theorem.

Published online 12 August 2026; First View · Source | DOI 10.1017/S0305004126102126

[106] Irreducibility of the characteristic polynomials of random tridiagonal matrices

Lior Bary-Soroker, Daniele Garzoni, Sasha Sodin. 2026. Journal of Number Theory 280, 973-998.

Under RH for certain Dedekind zeta functions, proves irreducibility with exponentially high probability and alternating-or-symmetric Galois group for a specified tridiagonal model. Also analyzes a related model with a smaller Galois group, illustrating why the matrix ensemble matters.

Published; conditional on RH · Source | DOI 10.1016/j.jnt.2025.09.008 | arXiv:2502.17218

Probabilistic permutation-group methods

[107] On random generation of the symmetric group

Tomasz Łuczak, László Pyber. 1993. Combinatorics, Probability and Computing 2(4), 505-512.

Supplies the permutation-group principle behind several probabilistic Galois arguments: a random permutation rarely belongs to a proper transitive subgroup other than A_n. Invariable generation is relevant because modular factorization determines conjugacy classes, rather than labeled permutations.

Published · Source | DOI 10.1017/S0963548300000869

[108] Permutations contained in transitive subgroups

Sean Eberhard, Kevin Ford, Dimitris Koukoulopoulos. 2016. Discrete Analysis 2016, 12.

Sharpens quantitative control of permutations lying in transitive subgroups other than S_n or A_n. Separates imprimitive and primitive obstructions, making it useful in arguments that first establish irreducibility and then force a large Galois group.

Published · Source | DOI 10.19086/da.849 | arXiv:1605.01068

[109] Invariable generation of the symmetric group

Sean Eberhard, Kevin Ford, Ben Green. 2017. Duke Mathematical Journal 166(8), 1573-1590.

Shows that three uniform random permutations fail to invariably generate S_n with probability tending to one, whereas four succeed with probability bounded away from zero. Explains why information from a small fixed collection of modular factorizations has a nontrivial combinatorial threshold.

Published · Source | DOI 10.1215/00127094-0000007X | arXiv:1508.01870

06 Hilbert irreducibility and inverse Galois theory

Hilbert irreducibility and specialization

[110] Ueber die Irreducibilität ganzer rationaler Functionen mit ganzzahligen Coefficienten

David Hilbert. 1892. Journal für die reine und angewandte Mathematik 110, 104-129.

The original source for Hilbert's irreducibility theorem. It is the historical starting point for passing from irreducible polynomial families over a rational function field to irreducible arithmetic specializations; use a modern treatment for the general language of Hilbertian fields.

Published article · Source | DOI 10.1515/crll.1892.110.104

[111] Fields containing values of algebraic functions

Roberto Dvornicich, Umberto Zannier. 1994. Annali della Scuola Normale Superiore di Pisa, Classe di Scienze (4) 21(3), 421-443.

Studies the number fields generated by specialized values of algebraic functions. A useful bridge from the existence of irreducible specializations to questions about the diversity of the fields produced.

Published article · Source

[112] Specializations of one-parameter families of polynomials

Farshid Hajir, Siman Wong. 2006. Annales de l’Institut Fourier 56(4), 1127-1163.

Develops geometric and group-theoretic conditions that make the exceptional set of reducible specializations over a number field finite. For fixed n ≥ 10, all but finitely many parameter values in any fixed number field give irreducible generalized Laguerre polynomials with Galois group S_n.

Published article · Source | DOI 10.5802/aif.2208 | arXiv:math/0405139

[113] Cyclotomic Diophantine problems (Hilbert irreducibility and invariant sets for polynomial maps)

Roberto Dvornicich, Umberto Zannier. 2007. Duke Mathematical Journal 139(3), 527-554.

Develops Hilbert irreducibility in cyclotomic extensions using the arithmetic of torsion points on tori. Relevant when specialization parameters are roots of unity or when polynomial dynamics is restricted to cyclotomic fields; its structural hypotheses must be retained.

Published article · Source | DOI 10.1215/S0012-7094-07-13934-6

[114] Topics in Galois Theory

Jean-Pierre Serre. 2007. A K Peters/CRC Press, Research Notes in Mathematics 1, ed. 2.

A compact route from Hilbert irreducibility and thin sets to explicit Galois realizations, rigidity, and sieve methods. Chapter 3 and Section 4.6 are particularly useful for explaining why generic Galois groups survive many specializations.

Published monograph · Source | DOI 10.1201/b10588

[115] Hilbert irreducibility above algebraic groups

Umberto Zannier. 2010. Duke Mathematical Journal 153(2), 397-425.

Studies irreducible fibers above Zariski-dense cyclic subgroups of algebraic tori and powers of an elliptic curve without complex multiplication, under a pullback irreducibility condition. The method combines torsion specialization with descent using Chebotarev, and yields a Bertini-type result for tori.

Published article · Source | DOI 10.1215/00127094-2010-027 | arXiv:1001.3079

[116] Specialization results in Galois theory

Pierre Dèbes, François Legrand. 2013. Transactions of the American Mathematical Society 365(10), 5259-5275.

Uses twisting to turn prescribed specialization questions into rational-point questions. One application combines irreducible specialization with prescribed residue-degree partitions at finitely many sufficiently large primes for covers over Q with geometric monodromy S_n.

Published article · Source | DOI 10.1090/S0002-9947-2013-05800-X | arXiv:1106.6151

[117] Field Arithmetic

Michael D. Fried, Moshe Jarden. 2023. Springer, Ergebnisse der Mathematik und ihrer Grenzgebiete, 3. Folge, 11, ed. 4.

A major reference for Hilbertian fields, absolute Galois groups, specialization, and PAC fields. The fourth edition updates the earlier editions and includes a new chapter on Hilbertian subfields of Galois extensions.

Published monograph · Source | DOI 10.1007/978-3-031-28020-7

Generic polynomials and inverse Galois theory

[118] Groups as Galois Groups: An Introduction

Helmut Völklein. 1996. Cambridge Studies in Advanced Mathematics 53.

Introduces the geometric approach to inverse Galois theory through covers, Riemann's existence theorem, and rigidity. Especially useful for learning how permutation data can be converted into polynomial families with a desired group.

Published monograph · Source | DOI 10.1017/CBO9780511471117

[119] Generic Polynomials: Constructive Aspects of the Inverse Galois Problem

Christian U. Jensen, Arne Ledet, Noriko Yui. 2002. Mathematical Sciences Research Institute Publications 45.

A constructive source for generic polynomials and the invariant-theoretic methods behind them. The freely accessible MSRI/SLMath page supplies downloadable chapters and concrete families for specified finite groups.

Published monograph · Source

[120] Inverse Galois Theory

Gunter Malle, B. Heinrich Matzat. 2018. Springer Monographs in Mathematics, ed. 2.

A systematic reference for constructing Galois extensions through rigidity, braid actions, embedding problems, additive polynomials, and rigid analytic methods. It also includes tables of example polynomials for transitive and primitive groups of small degree.

Published monograph · Source | DOI 10.1007/978-3-662-55420-3

07 Composition, iteration, and arboreal groups

Iteration, composition, and arboreal Galois groups

[121] The Galois Theory of Iterates and Composites of Polynomials

R. W. K. Odoni. 1985. Proceedings of the London Mathematical Society (3) 51(3), 385-414.

A foundational treatment of Galois groups under composition and iteration. In characteristic zero, the iterates of the generic degree-d polynomial have the full iterated wreath-product group, leading to finite-level specialization results and the later arboreal questions.

Published article · Source | DOI 10.1112/plms/s3-51.3.385

[122] Galois groups over Q of some iterated polynomials

Michael Stoll. 1992. Archiv der Mathematik 59, 239-244.

Determines maximality of iterated quadratic Galois groups through arithmetic square-class conditions. A useful concrete model for how critical-orbit information controls the growth of splitting fields under iteration.

Published article · Source | DOI 10.1007/BF01197321

[123] Irreducibility of the iterates of a quadratic polynomial over a field

Mohamed Ayad, Donald L. McQuillan. 2000. Acta Arithmetica 93(1), 87-97.

Develops criteria for irreducibility of successive quadratic iterates. Read it together with the authors' 2001 correction before using the stated tests in a proof or computation.

Published article; correction available · Source | DOI 10.4064/aa-93-1-87-97

[124] Corrections to "Irreducibility of the iterates of a quadratic polynomial over a field" (Acta Arith. 93 (2000), 87-97)

Mohamed Ayad, Donald L. McQuillan. 2001. Acta Arithmetica 99(1), 97.

The published correction to Ayad and McQuillan's quadratic-iteration paper. Keep this record linked to the original article when building a methods database.

Published correction · Source | DOI 10.4064/aa99-1-9

[125] An iterative construction of irreducible polynomials reducible modulo every prime

Rafe Jones. 2012. Journal of Algebra 369, 114-128.

Constructs polynomials of arbitrarily large degree that are irreducible over Q, and over suitable other global fields, but reducible modulo every prime. Failure to find an irreducible reduction is therefore not evidence of reducibility over the ground field.

Published article · Source | DOI 10.1016/j.jalgebra.2012.05.020 | arXiv:1012.2857

[126] Galois representations from pre-image trees: an arboreal survey

Rafe Jones. 2013. Publications mathématiques de Besançon. Algèbre et théorie des nombres, 107-136.

A readable entry point to Galois actions on trees of iterated preimages, stability, and density questions. Use its bibliography for the foundational literature, then supplement its open-problem discussion with later work.

Published survey · Source | DOI 10.5802/pmb.a-154 | arXiv:1402.6018

[127] Polynomials with Surjective Arboreal Galois Representations Exist in Every Degree

Joel Specter. 2018. arXiv preprint.

Gives the number-field form of Odoni's existence conjecture in all degrees. The source checked is the 2018 preprint, cited as such in subsequent literature; this must be distinguished from the false assertion over all Hilbertian fields.

Preprint; version 2, 12 March 2018 · Source | arXiv:1803.00434

[128] Odoni's conjecture for number fields

Robert L. Benedetto, Jamie Juul. 2019. Bulletin of the London Mathematical Society 51(2), 237-250.

Proves the desired existence of full arboreal Galois groups for even degree over every number field and for odd degree when the number field has odd degree over Q. The title should not be read as asserting every number-field case within this paper alone.

Published article · Source | DOI 10.1112/blms.12225 | arXiv:1803.01987

[129] Iterates of Generic Polynomials and Generic Rational Functions

Jamie Juul. 2019. Transactions of the American Mathematical Society 371(2), 809-831.

Extends the generic full wreath-product result to arbitrary characteristic for rational functions and to polynomials except the degree-2, characteristic-2 case. The paper also records the contributions of Odoni's unpublished work to these results.

Published article · Source | DOI 10.1090/tran/7229 | arXiv:1410.3814

[130] Dynamical Galois groups of trinomials and Odoni's conjecture

Nicole R. Looper. 2019. Bulletin of the London Mathematical Society 51(2), 278-292.

Constructs integral polynomials with surjective arboreal representations in every prime degree. It also gives conditional results in certain composite degrees using Vojta's conjecture, so the unconditional and conditional statements should be separated.

Published article · Source | DOI 10.1112/blms.12227 | arXiv:1609.03398

[131] Odoni's conjecture on arboreal Galois representations is false

Philip Dittmann, Borys Kadets. 2022. Proceedings of the American Mathematical Society 150(8), 3335-3343.

Disproves Odoni's conjecture in its general formulation over Hilbertian fields of characteristic zero. It does not contradict positive existence results over number fields; the difference in the base-field quantifier is essential.

Published article · Source | DOI 10.1090/proc/15920 | arXiv:2012.03076

Composition, indecomposability and monodromy

[132] Prime and composite polynomials

Joseph F. Ritt. 1922. Transactions of the American Mathematical Society 23(1), 51-66.

The classical source for functional decomposition of complex polynomials into indecomposable components. Here prime means indecomposable under composition, not irreducible under multiplication; read together with the erratum in the same volume, p. 431.

Published article with erratum · Source | DOI 10.1090/S0002-9947-1922-1501189-9

[133] Primitive monodromy groups of polynomials

Peter Müller. 1995. Recent Developments in the Inverse Galois Problem, Contemporary Mathematics 186, 385-401.

Classifies geometric monodromy groups of indecomposable complex polynomials and discusses restrictions for polynomials with rational coefficients. The group is that of f(X)-t over a function field, rather than the splitting-field group of f(X) over the coefficient field.

Published book chapter; author-hosted version · Source

[134] On Ritt's polynomial decomposition theorems

Michael E. Zieve, Peter Müller. 2008. arXiv preprint.

Develops a structured description of all complete decompositions of a complex polynomial, refining the classical Ritt framework. Both authors' publication pages still list it as a preprint, so it should be cited with its arXiv identifier rather than an invented journal reference.

Preprint; no journal publication verified · Source | arXiv:0807.3578

[135] Jordan-Hölder theorem for imprimitivity systems and maximal decompositions of rational functions

Mikhail Muzychuk, Fedor Pakovich. 2011. Proceedings of the London Mathematical Society 102(1), 1-24.

Studies imprimitivity lattices for permutation groups with a cyclic subgroup having at most two orbits, with applications to decomposition. It extends aspects of Ritt's first theorem and explains why unrestricted rational-function analogues can fail.

Published article · Source | DOI 10.1112/plms/pdq009

08 Finite fields: construction and distribution

Finite fields: construction and prescribed coefficients

[136] New algorithms for finding irreducible polynomials over finite fields

Victor Shoup. 1990. Mathematics of Computation 54, 435-447.

Constructs irreducible polynomials of a prescribed degree, including deterministic polynomial-time results for small characteristic. Distinguish dependence polynomial in the characteristic from dependence polynomial in its bit length.

Published article; author-hosted version · Source | DOI 10.1090/S0025-5718-1990-0993933-0

[137] Primitive polynomials over finite fields

Tom Hansen, Gary L. Mullen. 1992. Mathematics of Computation 59(200), 639-643.

Provides explicit primitive-polynomial tables and formulates influential existence conjectures with one prescribed coefficient. The primitivity question and the irreducibility question are distinct; a primitive polynomial has a root generating the multiplicative group of the extension field.

Published article with supplementary tables · Source | DOI 10.1090/S0025-5718-1992-1134730-7

[138] Fast construction of irreducible polynomials over finite fields

Victor Shoup. 1994. Journal of Symbolic Computation 17, 371-391.

Develops faster algorithms for constructing an irreducible polynomial of specified degree over a finite field. Useful when the objective is an explicit extension-field representation rather than testing a single given polynomial.

Published article · Source | DOI 10.1006/jsco.1994.1025

[139] Generators and irreducible polynomials over finite fields

Daqing Wan. 1997. Mathematics of Computation 66(219), 1195-1212.

Uses character-sum estimates to study generators and irreducible polynomials with prescribed coefficients. Establishes the Hansen-Mullen irreducibility assertion outside finitely many cases; Ham and Mullen settle the remaining cases.

Published article · Source | DOI 10.1090/S0025-5718-97-00835-1

[140] Distribution of irreducible polynomials of small degrees over finite fields

Kie H. Ham, Gary L. Mullen. 1998. Mathematics of Computation 67(221), 337-341.

Completes the Hansen-Mullen irreducibility conjecture by verifying the finite set of cases left by Wan's theorem. The statement includes necessary exceptions, notably a prescribed zero constant coefficient and the quadratic zero-linear-coefficient case in characteristic two.

Published article · Source | DOI 10.1090/S0025-5718-98-00904-1

[141] Primitive polynomials with a prescribed coefficient

Stephen D. Cohen. 2006. Finite Fields and Their Applications 12(3), 425-491.

Establishes the Hansen-Mullen primitivity assertion for degrees at least nine by character-sum and sieve methods. This is a stronger condition than irreducibility and this paper alone is not the full treatment of the small-degree cases.

Published article · Source | DOI 10.1016/j.ffa.2005.08.001

[142] Handbook of Finite Fields

Gary L. Mullen, Daniel Panario (eds.). 2013. CRC Press.

A broad reference for irreducible and primitive polynomials, prescribed coefficients, factorization algorithms, and finite-field applications. Its extensive bibliography is a practical starting point for extending this guide in characteristic p.

Published edited handbook · Source | DOI 10.1201/b15006

[143] Irreducible polynomials with several prescribed coefficients

Paul Pollack. 2013. Finite Fields and Their Applications 22, 70-78.

Gives an explicit estimate for monic irreducibles with coefficients prescribed at arbitrary positions, with a nonzero constant term when it is prescribed. For sufficiently large degree n, it allows about $(1-\varepsilon)\sqrt{n}$ prescribed coefficients uniformly over the ground finite field.

Published article; author-hosted version · Source | DOI 10.1016/j.ffa.2013.03.001

[144] Irreducible polynomials with several prescribed coefficients

Junsoo Ha. 2016. Finite Fields and Their Applications 40, 10-25.

Improves prescribed-coefficient results to a positive proportion of the degree by adapting Bourgain's method. Its large-field range approaches one quarter of the coefficients; the characteristic, size, and nonzero-constant-term conditions must be retained when applying the theorem.

Published article · Source | DOI 10.1016/j.ffa.2016.02.006 | arXiv:1601.06867

[145] A new proof of the Hansen-Mullen irreducibility conjecture

Aleksandr Tuxanidy, Qiang Wang. 2018. Canadian Journal of Mathematics 70(6), 1373-1389.

Gives a different proof of the one-prescribed-coefficient existence theorem using least periods of discrete Fourier transforms over finite fields. A useful example of a method that differs from the character-sum approach while preserving the genuine exceptional cases.

Published article · Source | DOI 10.4153/CJM-2017-022-1 | arXiv:1604.04023

Finite fields: factorization statistics and monodromy

[146] Factorization of polynomials over finite fields

Richard G. Swan. 1962. Pacific Journal of Mathematics 12(3), 1099-1106.

Presents the Stickelberger discriminant-parity theorem, including characteristic two via lifting, and derives reducibility obstructions for binary trinomials. Swan explicitly credits the basic parity theorem to Stickelberger and notes Dalen's earlier proof.

Published article · Source | DOI 10.2140/pjm.1962.12.1099

[147] The distribution of polynomials over finite fields

Stephen D. Cohen. 1970. Acta Arithmetica 17(3), 255-271.

A foundational study of the factorization patterns of f(X)-a g(X) as a varies over a finite field. It connects prescribed factor degrees to permutation cycle types in a function-field Galois group, with the arithmetic hypotheses governing the distribution.

Published article · Source | DOI 10.4064/aa-17-3-255-271

[148] Polynomials with PSL(2) monodromy

Robert M. Guralnick, Michael E. Zieve. 2010. Annals of Mathematics 172(2), 1315-1359.

Classifies positive-characteristic polynomials with specified PSL(2) monodromy under a ramification hypothesis, with applications to exceptional polynomials and decomposition after extending constants. This concerns function-field Galois groups; splitting fields of individual polynomials over finite fields have cyclic Galois groups.

Published article · Source | DOI 10.4007/annals.2010.172.1315 | arXiv:0707.1835

[149] Prime polynomials in short intervals and in arithmetic progressions

Efrat Bank, Lior Bary-Soroker, Lior Rosenzweig. 2015. Duke Mathematical Journal 164(2), 277-295.

Proves asymptotic counts for irreducibles in short polynomial intervals and arithmetic progressions, and treats other factorization types. The asymptotic is fixed degree with the field size tending to infinity, with explicit small-characteristic and interval-length restrictions.

Published article · Source | DOI 10.1215/00127094-2856728 | arXiv:1302.0625

[150] Factorization statistics of restricted polynomial specializations over large finite fields

Alexei Entin. 2021. Israel Journal of Mathematics 242(1), 37-53.

Studies factorization patterns when specialization parameters are restricted to sufficiently large, regular subsets of a prime field. The degree stays fixed as the characteristic tends to infinity; the result is not a growing-degree statement over one fixed finite field.

Published article · Source | DOI 10.1007/s11856-021-2101-9 | arXiv:1810.07512

09 Algorithms for factorization and Galois groups

finite fields and factorization

[151] Factoring polynomials over finite fields

Elwyn R. Berlekamp. 1967. Bell System Technical Journal 46(8), 1853-1859.

The foundational linear-algebra method for factoring polynomials over finite fields. Read it to understand the Frobenius-fixed subalgebra that detects the irreducible factors.

Published article · Source | DOI 10.1002/j.1538-7305.1967.tb03174.x

[152] On Hensel factorization, I

Hans Zassenhaus. 1969. Journal of Number Theory 1, 291-311.

The classical modular route to integer-polynomial factorization: factor modulo a prime, lift p-adically, then recombine the lifted factors. It explains why local factors must be grouped before recovering rational factors.

Published article · Source | DOI 10.1016/0022-314X(69)90047-X

[153] Probabilistic algorithms in finite fields

Michael O. Rabin. 1980. SIAM Journal on Computing 9(2), 273-280.

A central source for finite-field algorithms for irreducible-polynomial construction, root finding, and factorization. Useful alongside the exact Frobenius-power criterion for testing a monic polynomial for irreducibility.

Published article · Source | DOI 10.1137/0209024

[154] A new algorithm for factoring polynomials over finite fields

David G. Cantor, Hans Zassenhaus. 1981. Mathematics of Computation 36(154), 587-592.

A basic randomized finite-field factorization algorithm, especially important for splitting products whose irreducible factors have equal degree. Its randomness concerns the search procedure; successful factors can be checked exactly.

Published article · Source | DOI 10.1090/S0025-5718-1981-0606517-5

[155] Factoring polynomials with rational coefficients

A. K. Lenstra, Jr., H. W. Lenstra, L. Lovász. 1982. Mathematische Annalen 261(4), 515-534.

Proves polynomial-time factorization over the rationals using lattice basis reduction. The LLL algorithm introduced here is also a central tool for finding algebraic relations and constructing certificates.

Published article · Source | DOI 10.1007/BF01457454

[156] A new polynomial factorization algorithm and its implementation

Victor Shoup. 1995. Journal of Symbolic Computation 20(4), 363-397.

Develops a practical implementation of baby-step/giant-step finite-field factorization. A useful bridge between asymptotic algorithms and actual computational work.

Published article · Source | DOI 10.1006/jsco.1995.1055

[157] Subquadratic-time factoring of polynomials over finite fields

Erich Kaltofen, Victor Shoup. 1998. Mathematics of Computation 67(223), 1179-1197.

A major complexity advance in randomized polynomial factorization over finite fields. Its baby-step/giant-step framework is part of the background to modern finite-field factorization libraries.

Published article · Source | DOI 10.1090/S0025-5718-98-00944-2

[158] Factoring polynomials and the knapsack problem

Mark van Hoeij. 2002. Journal of Number Theory 95(2), 167-189.

Transforms the expensive recombination stage of modular factorization into a structured knapsack problem handled by lattice reduction. This is one of the main sources behind practical rational-polynomial factorization.

Published article · Source | DOI 10.1006/jnth.2001.2763

[159] Factoring polynomials over global fields

Karim Belabas, Mark van Hoeij, Jürgen Klüners, Allan Steel. 2009. Journal de théorie des nombres de Bordeaux 21(1), 15-39.

Establishes polynomial-time complexity for van Hoeij's original rational factorization algorithm and natural variants. It also treats global-field extensions of the approach and bivariate factorization over finite fields.

Published article · Source | DOI 10.5802/jtnb.655

Galois groups and resolvents

[160] The determination of Galois groups

R. P. Stauduhar. 1973. Mathematics of Computation 27(124), 981-996.

Introduces the subgroup-descent strategy based on relative invariants and resolvents. This remains the conceptual starting point for many practical Galois-group computations.

Published article · Source | DOI 10.1090/S0025-5718-1973-0327712-4

[161] Solvability by radicals is in polynomial time

Susan Landau, Gary Lee Miller. 1985. Journal of Computer and System Sciences 30(2), 179-208.

Shows that deciding solvability by radicals for rational polynomials is possible in polynomial time. This decision problem should be distinguished from constructing an entire splitting field or listing every element of a large Galois group.

Published article · Source | DOI 10.1016/0022-0000(85)90013-3

[162] Computing Galois groups over the rationals

Leonard Soicher, John McKay. 1985. Journal of Number Theory 20(3), 273-281.

Develops practical computational methods using resolvents and low-degree transitive groups. The paper also gives rational realizations of the transitive permutation groups in degrees three through seven.

Published article · Source | DOI 10.1016/0022-314X(85)90022-8

[163] Solving solvable quintics

David S. Dummit. 1991. Mathematics of Computation 57(195), 387-401.

A concrete sextic-resolvent treatment of solvable irreducible quintics and radical expressions. Use it with Robin Chapman's 1992 correction to the proof of Theorem 1 and the numerical errata on the author's webpage.

Published article; corrections available · Source | DOI 10.1090/S0025-5718-1991-1079014-X

[164] Galois group computation for rational polynomials

Katharina Geissler, Jürgen Klüners. 2000. Journal of Symbolic Computation 30(6), 653-674.

A practical development of Galois-group computation using p-adic roots and refinements of Stauduhar's method. Read it before the later degree-independent approach of Fieker and Klüners.

Published article · Source | DOI 10.1006/jsco.2000.0377

[165] Explicit Galois realization of transitive groups of degree up to 15

Jürgen Klüners, Gunter Malle. 2000. Journal of Symbolic Computation 30(6), 675-716.

Constructs explicit polynomials realizing all transitive groups of degree at most fifteen over the rationals. A valuable source of test examples and methods for organizing computations by permutation group.

Published article · Source | DOI 10.1006/jsco.2000.0378

[166] Identifying Frobenius elements in Galois groups

Tim Dokchitser, Vladimir Dokchitser. 2013. Algebra & Number Theory 7(6), 1325-1352.

Identifies Frobenius conjugacy classes more precisely than the factor-degree data of reduction modulo a prime. Useful when distinct conjugacy classes have the same cycle type in the chosen permutation representation.

Published article · Source | DOI 10.2140/ant.2013.7.1325 | arXiv:1009.5388

[167] Computation of Galois groups of rational polynomials

Claus Fieker, Jürgen Klüners. 2014. LMS Journal of Computation and Mathematics 17(1), 141-158.

Constructs relative invariants for practical Galois-group computation without an a priori degree bound. It is a central reference for understanding current Magma computations; practical time and memory limits still matter.

Published article · Source | DOI 10.1112/S1461157013000302 | arXiv:1211.3588

reference books

[168] A Course in Computational Algebraic Number Theory

Henri Cohen. 1993. Springer, Graduate Texts in Mathematics 138.

A systematic computational number-theory reference, including polynomial factorization, number fields, and Galois-group computations. Particularly helpful for translating a mathematical method into a reproducible algorithm.

Published book · Source | DOI 10.1007/978-3-662-02945-9

[169] Modern Computer Algebra

Joachim von zur Gathen, Jürgen Gerhard. 2013. Cambridge University Press, ed. 3.

A comprehensive source on polynomial arithmetic, finite-field factorization, Hensel lifting, and factorization over the rationals. Use it for complexity models and complete algorithmic proofs rather than isolated software recipes.

Published book · Source | DOI 10.1017/CBO9781139856065

10 Databases and software

[170] A database for field extensions of the rationals

Jürgen Klüners, Gunter Malle. 2001. LMS Journal of Computation and Mathematics 4, 182-196.

Explains the arithmetic database, its construction methods, and its emphasis on Galois groups and signatures. Consult the live database separately, since its coverage has grown beyond the original paper.

Published article · Source | DOI 10.1112/S1461157000000851 | arXiv:math/0102232

[171] FLINT documentation: polynomial factorization algorithms

The FLINT Team. undated.

Implementation-oriented explanations of finite-field and integer-polynomial factorization, linked to the original algorithms. Especially useful for connecting Berlekamp, Cantor-Zassenhaus, Hensel lifting, and lattice-based recombination with executable library routines.

Maintained software documentation · Source

[172] A database for number fields

Jürgen Klüners, Gunter Malle. undated.

A source of defining polynomials indexed by Galois group and signature. At access, the site reports coverage through degree twenty-three for all transitive groups except M23, with stated limitations on signatures and discriminant minima.

Maintained mathematical database · Source

[173] The L-functions and Modular Forms Database: number fields and Galois groups

The LMFDB Collaboration. undated.

Searchable number-field records and transitive-group data, with labels, invariants, and links between objects. Consult the completeness and reliability pages for each collection; a group appearing in a transitive-group catalogue does not by itself prove its realization over the rationals.

Maintained mathematical database · Source

[174] Magma Handbook: Galois groups

Magma Computational Algebra System. undated.

Official documentation for the Stauduhar-based GaloisGroup implementation, with no a priori degree bound. The handbook explicitly directs users to GaloisProof to certify the conditional steps of GaloisGroup; record the successful certification with a research computation.

Maintained software documentation · Source

[175] PARI/GP reference manual: general number fields and polgalois

The PARI Group. undated.

Official reference for polgalois and number-field computations. The documented polgalois input must be irreducible and has degree at most seven, or eleven with galdata; the manual also warns about sensitivity to initial numerical precision.

Maintained software documentation · Source

[176] SageMath reference manual: number fields and Galois groups

The Sage Developers. undated.

Documents the number-field galois_group interface and explicit actions on a Galois closure. Backend choice matters: current documentation lists PARI, GAP, and optional Magma, so a Sage wrapper is not an independent second proof of the backend's result.

Maintained software documentation · Source

11 Multivariate absolute irreducibility

[177] Reduzibilität ebener Kurven

Wolfgang M. Ruppert. 1986. Journal für die reine und angewandte Mathematik 369, 167-191.

A classical source on reducibility of plane curves and algebraic criteria for absolute irreducibility. This is a multivariate question: a univariate polynomial of degree greater than one always splits over an algebraic closure.

Published article · Source | DOI 10.1515/crll.1986.369.167

Bibliographic details checked in Lecerf's survey and an AMS bibliography; the publisher full text was not retrieved.

[178] Effective Noether irreducibility forms and applications

Erich Kaltofen. 1995. Journal of Computer and System Sciences 50(2), 274-295.

Gives effective bounds for algebraic irreducibility criteria and applications to specialization and factorization complexity. Useful when one wants quantitative control of exceptional specializations in several variables.

Published article · Source | DOI 10.1006/jcss.1995.1023 | Open copy

[179] Absolute irreducibility of polynomials via Newton polytopes

Shuhong Gao. 2001. Journal of Algebra 237(2), 501-520.

Uses integral indecomposability of Newton polytopes to obtain absolute irreducibility criteria for sparse multivariate polynomials. Monomial factors must be excluded when applying a polytope criterion to a polynomial.

Published article · Source | DOI 10.1006/jabr.2000.8586

[180] Factoring multivariate polynomials via partial differential equations

Shuhong Gao. 2003. Mathematics of Computation 72(242), 801-822.

A factorization method based on a differential equation and linear algebra, in characteristic zero or sufficiently large characteristic. Connects rational and absolute factorization and provides an effective reduction from multivariate to bivariate problems.

Published article · Source | DOI 10.1090/S0025-5718-02-01428-X | Open copy

[181] Factorisation des polynômes à plusieurs variables

Grégoire Lecerf. 2013. Journées Nationales de Calcul Formel (13-17 May 2013), Les cours du CIRM(1), 1-85.

An extensive French-language course on multivariate factorization, with a substantial bibliography. A good continuation when the atlas expands from univariate arithmetic to absolute factorization and computational algebraic geometry.

Published lecture notes · Source | DOI 10.5802/ccirm.18

12 Free expositions and course materials

[182] Math 788F: The Theory of Irreducible Polynomials

Michael Filaseta. 2016. University of South Carolina course materials.

A focused course hub with a syllabus, problems, and a Newton polygon introduction. It is a useful model for learning by proof techniques; the page does not currently provide every chapter of the proposed book as an open link.

Expository notes · Source

[183] An irreducible that factors modulo all primes

Keith Conrad. undated. Author-hosted expository note.

Explains why irreducibility over the rationals need not be detectable by one irreducible reduction. The obstruction is visible in the cycle types available in the Galois group.

Expository notes · Source

[184] Galois groups of cubics and quartics (not in characteristic 2)

Keith Conrad. undated. Author-hosted expository note.

Worked low-degree classifications using discriminants and resolvents. The characteristic restriction matters; the author’s collection also contains a separate all-characteristics treatment.

Expository notes · Source

[185] Irreducibility tests in F_p[T]

Keith Conrad. undated. Author-hosted expository note.

An accessible account of irreducibility tests over finite fields. Useful for implementing transparent modular certificates rather than reporting an unexplained software answer.

Expository notes · Source

[186] Irreducibility tests in Q[T]

Keith Conrad. undated. Author-hosted expository note.

Clear proofs of Gauss’s lemma, reduction modulo a prime, and Eisenstein’s criterion. Good first reading before more specialized irreducibility criteria.

Expository notes · Source

[187] Galois groups over Q and factorizations mod p

Keith Conrad. undated. Author-hosted expository note.

A short explanation of how squarefree factorizations modulo good primes produce cycle types in a Galois group. Read the discriminant condition carefully.

Expository notes · Source

[188] Recognizing Galois groups S_n and A_n

Keith Conrad. undated. Author-hosted expository note.

Useful group-theoretic recognition criteria and examples. A practical bridge from modular factorization and real roots to identifying the full permutation group.

Expository notes · Source